

The Wang Method Series

Notebook #1 (Version 1.0)

Understanding the Top-Layer Four-Edge-Pair Patterns

Unusual Top-Layer Four-Edge-Pair Patterns After OLL Parity Correction

Laung-Terng (LT) Wang

August 1, 2026

1. Motivation

While solving several 4×4 Rubik's Cubes using the Yau Method followed by reduction, I repeatedly observed unusual top-layer four-edge-pair patterns after correcting the OLL parity and completing the remaining standard 3×3 OLL algorithms.

Instead of encountering only the familiar PLL parity cases, the cube occasionally reached several apparently abnormal four-edge-pair configurations. These patterns did not resemble the standard opposite-edge PLL parity cases commonly encountered during 4×4 cube solving.

These unexpected observations motivated a closer investigation into the relationship between these unusual patterns and the well-known PLL parity.

This notebook records observations made during actual cube solving. The experiments are exploratory and were originally conducted to understand these puzzling patterns. The observations presented here are independent of the particular OLL parity correction algorithm used.

The three experiments presented in this notebook illustrate three representative classes of illegal top-layer four-edge-pair configurations that were later found to account for all **ten illegal states** described in **Notebook #5**.

2. Initial Observation

After correcting the OLL parity to restore the yellow cross and subsequently applying standard 3×3 algorithms (such as *Sune*, *A-Perm*, and *Y-Perm*), all corners were solved. However, the four top-layer edge pairs occasionally remained in apparently abnormal configurations.

These configurations appeared unrelated to the familiar opposite-edge PLL parity cases.

This naturally raised the following question:

Are these unusual patterns genuinely new parity cases, or are they simply different visible representations of the same underlying PLL parity?

The following three representative experiments were designed to investigate this question.

3. Experiment 1 — A Zero-Aligned Illegal State

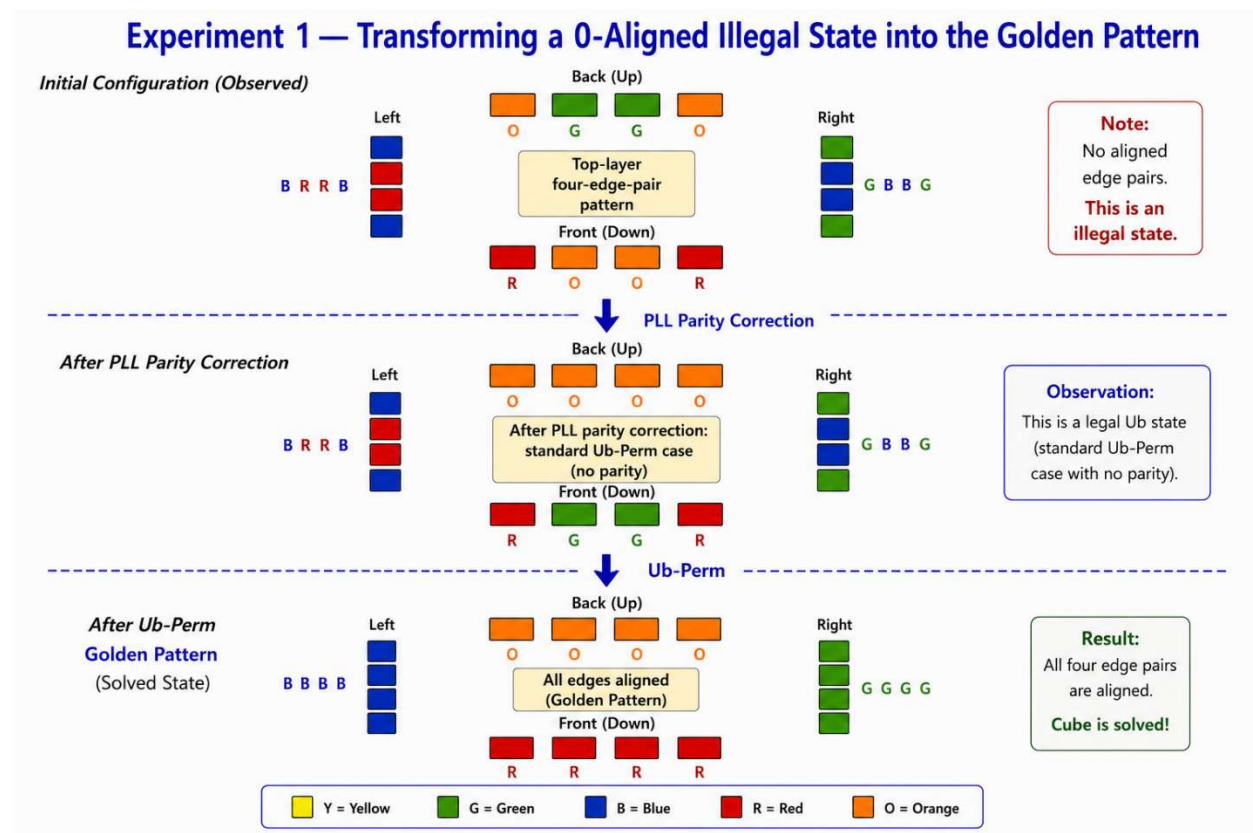


Figure 1.1. A zero-aligned illegal state.

Observed sequence is illustrated in Figure 1.1.

Observation

Applying the standard PLL parity correction transformed the illegal state into an ordinary 3×3 *Ub-Perm* case.

This experiment demonstrates that one class of illegal four-edge-pair configurations can be interpreted as another representation of the underlying PLL parity.

4. Experiment 2 — An Illegal State Transformed by *Z-Perm*

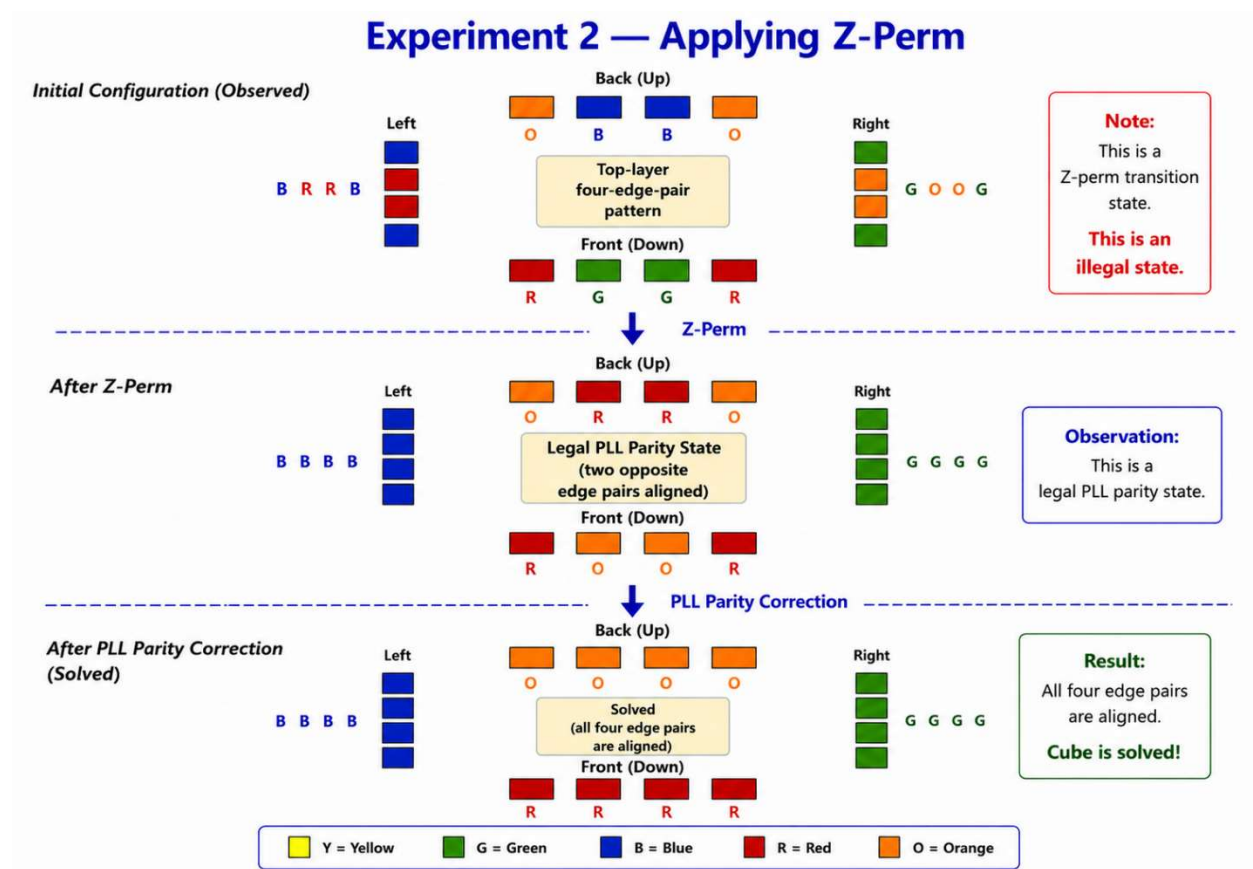


Figure 1.2. An illegal state transformed by *Z-Perm*.

Observed sequence is illustrated in Figure 1.2.

Observation

Applying *Z-Perm* transformed the illegal configuration into a familiar legal opposite-edge PLL parity state. After the standard PLL parity correction, the cube was solved immediately without requiring any additional PLL algorithm.

5. Experiment 3 — A Two-Adjacently-Aligned Illegal State

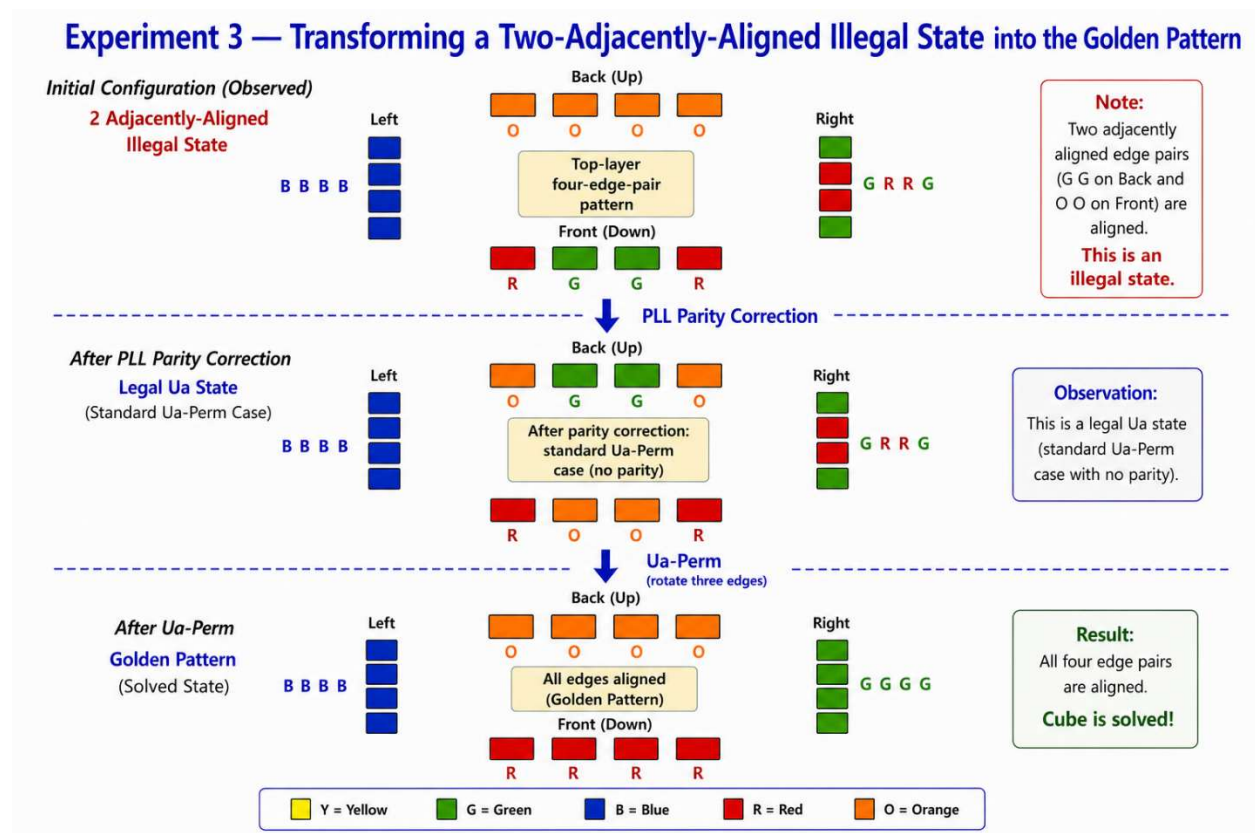


Figure 1.3. A two-adjacently-aligned illegal state.

Observed sequence is illustrated in Figure 1.3.

Observation

Applying the standard PLL parity correction transformed this illegal configuration into an ordinary 3×3 *Ua-Perm* case.

This experiment shows that another seemingly unrelated illegal configuration also corresponds to the same underlying parity phenomenon.

Together, Experiments 1–3 illustrate three representative classes of illegal four-edge-pair configurations.

6. Hypothesis

The observed illegal four-edge-pair configurations do not represent new types of parity.

Instead, they appear to be different visible representations of the same underlying PLL parity.

Standard 3×3 PLL algorithms, such as *Z-Perm*, do not eliminate the underlying parity. Instead, they may transform one visible representation of the parity into another.

A PLL parity correction removes the underlying parity and may leave either a legal *Ua-Perm* or a legal *Ub-Perm* case, depending on which opposite edge pair is selected during the correction. The two choices are symmetric.

7. Discussion

The three experiments consistently suggest that apparently different illegal four-edge-pair configurations may share the same underlying parity.

During the reduction phase of a 4×4 cube, wide-layer moves allow parity states to be created or eliminated.

Once reduction is complete, however, ordinary 3×3 algorithms appear to preserve the underlying parity while changing only its visible representation.

These observations eventually led to the broader structural classification presented in **Notebook #5**, where all twenty-four top-layer four-edge-pair patterns are classified into six structural groups, and the ten illegal states are organized into two deterministic transition paths.

Although the present notebook is based on experimental observations rather than formal proofs, the results consistently support this interpretation and motivate the structural framework presented in **Notebook #5**.

8. Open Questions

Several interesting questions remain for future investigation.

- Can these transformations be explained using permutation-group or graph-theoretic analysis?
 - Can illegal configurations be recognized before entering the 3×3 solving phase?
-

9. Lessons Learned

- Not every unusual four-edge-pair pattern represents a new parity.
 - Different illegal configurations may represent the same underlying parity.
 - PLL parity correction may lead to either *Ua-Perm* or *Ub-Perm*, depending on which opposite edge pair is selected during the correction.
 - Ordinary 3×3 algorithms appear to change only the visible representation of the parity.
 - These experimental observations motivated the structural framework developed later in **Notebook #5**.
-

Engineering Perspective

Several seemingly unrelated four-edge-pair configurations were found to share a common underlying parity behavior. These experimental observations eventually led to a structural classification of all twenty-four top-layer four-edge-pair patterns and their deterministic state transitions. Although still exploratory, this engineering perspective provides a logical interpretation of several puzzling parity phenomena encountered during 4×4 cube solving.